



EI-SCORE

Explainable-Interpretable and Simple
Customized Overall Ranking Engine

User Guide

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Introduction

EI-SCORE is an interactive, browser-based application for building Explainable and Interpretable Composite Indicators using decision rules. It uses the Dominance-based Rough Set Approach (DRSA) as described in Corrente et al. (2026).

The tool is designed for researchers, analysts, and decision-makers who need to rank or classify a set of units (countries, companies, students, patients, etc.) evaluated on multiple criteria, and who require transparent, human-readable justifications for the classification/scoring assigned.

Key features

- **Classification problems** – assigning alternatives to ordered classes (e.g., low, medium, high)
- **Scoring problems** – evaluating and ranking alternatives by a continuous score
- **Rule induction** – inducing at-least ($\mathcal{R}^{\geq}$) and at-most ($\mathcal{R}^{\leq}$) decision rules from reference units
- **Full pipeline** – inducing a maximal and a minimal set of rules able to classify or score all units without any contradiction
- **New unit assignment** – select a maximal and minimal set of rules able to classify or score with no contradiction a new set of units
- **Rule reuse** – visualize saved rules and apply to a datasets
- **Missing value support** – automatic detection and handling missing data

Reference

Corrente S., Greco S., Słowiński R., Zappalà S. (2026). *An explainable and interpretable composite indicator based on decision rules*. Omega, 142, 103513.
<https://doi.org/10.1016/j.omega.2026.103513>

Greco S., Matarazzo B., Słowiński R. (2001). *Rough sets theory for multicriteria decision analysis*. European Journal of Operational Research, 129, 1-47.
[https://doi.org/10.1016/S0377-2217\(00\)00167-3](https://doi.org/10.1016/S0377-2217(00)00167-3)

Installation and Launch

EI-SCORE can be used in two ways: as a locally installed application or via a web browser without any installation.

Option A – Web access (no installation required)

The application is publicly available online. Simply open the following URL in your browser:

<https://ei-score.streamlit.app>

No Python or package installation is required.

Option B – Local installation

Requirements:

- Python 3.14.3 or later
- numpy==2.4.2, pandas==3.0.1, scipy==1.17.0, streamlit==1.58.0

Step 1 – Clone the repository:

```
git clone https://github.com/silvanozap/Explainable-composite-indicator.git
cd Explainable-composite-indicator
```

Step 2 – Install dependencies:

```
pip install -r requirements.txt
```

Step 3 – Launch the application:

```
streamlit run drsa/interface/app.py
```

The application opens automatically at <http://localhost:8501>.

Input File Formats

Units file – load units in EI-SCORE

The data file must be a CSV or TXT file with the following structure:

```
Name,criterion_1,criterion_2,...,criterion_k,class
A1,4,3,2,1
A2,5,4,3,2
A3,7,6,5,
A4,6,5,4,
```

- **First column** (optional): unit names. If the first column contains non-numeric values, it is automatically interpreted as unit names.
- **Criteria columns**: numeric values. Any column separators (comma, semicolon, tab, space) are supported and selectable in the interface.
- **Last column**: class/score label with increasing direction.
 - Numeric value → **reference unit** (used for rule induction)

– Empty / NaN → **non-reference unit**

- **Missing values:** empty cells in criteria columns are detected and handled automatically.

Tip

Sample data files covering all use cases are available in the **examples/** folder of the repository, and can also be downloaded directly from the application sidebar.

Warning

The class/score label has increasing direction

Rules file (classification) – Load previous classification rules induced

Rules exported from the application or created manually follow this format:

```
#directions,increasing,decreasing,...
#mode,class
type,class,criterion_1,criterion_2,...
at-least,2,0.762,,
at-most,1,,2.71,
```

- **#directions:** preference direction of each criterion (increasing or decreasing)
- **#mode:** class for classification problems
- **Data rows:** each row is a rule with **type** (at-least or at-most), **class** (assigned class), and threshold values for each criterion (empty if the criterion is not in the rule condition).

Rules file (scoring) – Load previous scoring rules induced

For scoring problems, the **#mode** line changes and class values represent score values:

```
#directions,increasing,decreasing,...
#mode,score
type,class,criterion_1,criterion_2,...
at-least,16.67,0.762,,
at-most,0,,2.71,
```

Tip

Sample rules files for both classification and scoring can be downloaded directly from Tab 5 (Apply Rules) of the application.

Tab 1 – Data & Setup

The Data & Setup tab is the starting point of every analysis. It allows the user to load data, configure the problem, and set all analysis parameters.

Data

Upload CSV or TXT

PTF.csv 2.9KB

Separator

Quick start

1. **Data & Setup** — upload dataset, set directions and settings
2. **Run** — induce only rules or full pipeline
3. **Assignment** — inspect assignment of all units
4. **New Units** — assign new units with previous induced rules
5. **Apply Rules** — load saved rules and assign units

User guide

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EI-SCORE

Explainable-Interpretable and Simple Customized Overall Ranking Engine

User-friendly GUI to build your customized composite indicator based on Decision Rules

Data & Setup Run Assignment New Units Apply Rules

Dataset

	Code	Return	Beta	Net_Profit_Margin	ROA	ROE	EPS	P_E	P_BV	class
0	A1	86.692	1.146	10.327	4.94	22.855	402	7.169	2.183	2
1	A2	23.666	0.322	31.513	21.389	29.745	9331.333	6.701	1.905	None
2	A3	24.695	1.794	2.418	2.644	9.642	127.333	16.885	1.007	None
3	A4	24.145	1.489	30.193	6.101	21.223	469.667	11.951	1.742	3
4	A5	0.279	1.42	-1.236	-0.081	-39.005	-58.333	100	3.051	None
5	A6	103.495	1.379	12.536	14.722	59.18	2603.667	7.738	2.981	None

Reference units

Select reference units

A1 x A4 x A12 x A17 x A18 x A26 x A33 x A36 x A45 x

Total units: 50 Reference units: 9 Non-reference units: 41

Figure 1: Tab 1 – Data & Setup

Upload data

Click the **Upload CSV or TXT** button in the sidebar to load your data file. Select the appropriate column separator (comma, semicolon, tab, or space) from the dropdown. Once uploaded, the dataset is displayed in the main area.

Reference units

Reference units are the ones whose class/score assignments are known and are used for rule induction. By default, all units with a non-empty class/score label (last column of the loaded dataset) are considered reference units. You can manually select a subset of reference units using the multiselect widget.

Tip

In Tab 2 (Run) non-reference units do not participate in *rule induction only* but participate in *full pipeline*.

Preference directions of criteria

For each criterion, select the preference direction:

- **Increasing** (↑) (default) – higher values are preferred
- **Decreasing** (↓) – lower values are preferred

Criteria preference direction

Return: ↑ Increasing Beta: ↑ Increasing Net_Profit_Margin: ↑ Increasing ROA: ↑ Increasing ROE: ↑ Increasing EPS: ↑ Increasing

P_E: ↑ Increasing P_BV: ↑ Increasing

Settings

Min confidence (c):

1.00

Random seed: 1 - +

Figure 2: Preference direction of criteria selection

Settings

- **Min confidence (c) (default: 1)**: minimum confidence threshold for rule induction. Reducing this value allows the induction of approximate rules.
- **Random seed (default: 1)**: controls the random number generator used in the greedy selection step in case of running *full pipeline* in Tab 2 (Run).

Tab 2 – Run

The Run tab is where *rule induction* and the *full pipeline* are executed.

Figure 3: Tab 2 – Rule induction only

Problem type

Select the problem type using the **Problem type** radio button:

- **Classification (default)** – the last column represents a classification (integer number)
- **Scoring** – the last column represents scores (continuous number).

Rule induction only

Select **Rule induction only**: induction of at-least ($\mathcal{R}^{\geq}$) and at-most ($\mathcal{R}^{\leq}$) decision rules from the reference units.

The induced rules are displayed in collapsible expander, showing:

- Rule text with condition and decision assignment
- Support value (fraction of reference units covered)
- Matching units (units satisfying the rule condition)

Warning

The induced rules may provide contradictory classification for non-reference units in the Assignment tab.

Full pipeline

Select **Full pipeline** to run the rule induction avoiding any contradictory assignment:

1. **Step 1:** Rule induction from reference units
2. **Step 2:** Greedy selection of rules that avoid contradictory assignment of all units
3. **Step 3:** Assignment of all units using the induced rules
4. **Step 4:** Rule induction from all units to create the maximal set of rules
5. **Step 5:** MILP-based minimization to extract a minimal set of rules

A progress bar shows the pipeline execution. Upon completion, a summary table shows the number of rules at each step, maximal and minimal set of rules.

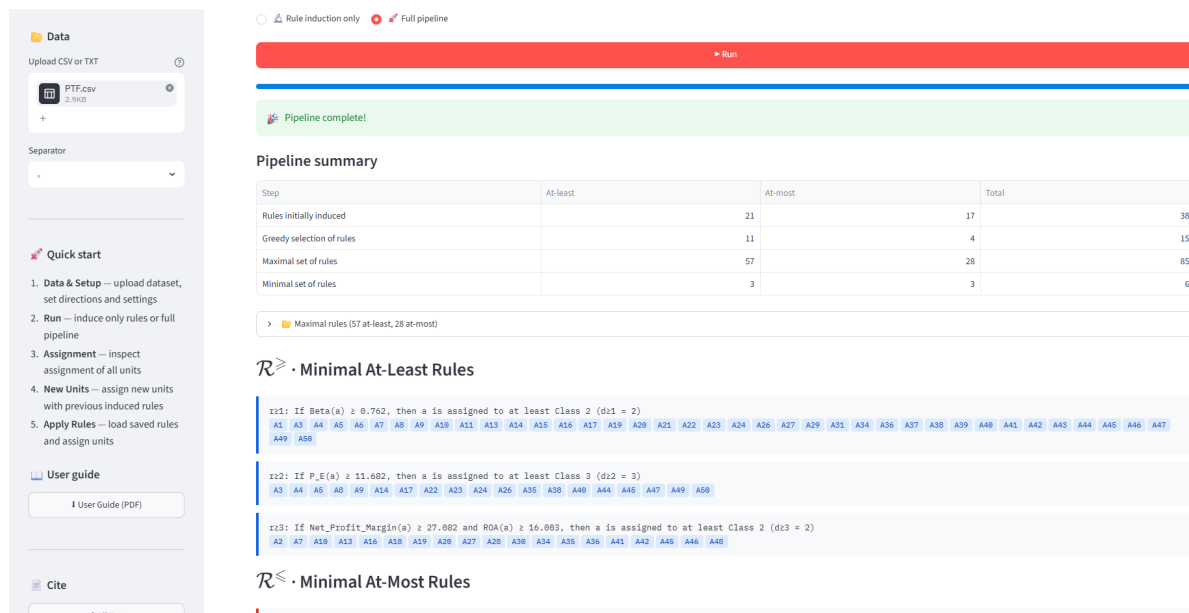


Figure 4: Tab 2 – Full pipeline summary

Exporting rules

After running, the following CSV files can be downloaded:

- **Maximal rules CSV:** the full set of rules
- **Minimal rules CSV:** a minimal set of rules

Tip

The download CSV can be used to load rules in Tab 5 (Apply rules).

Tab 3 – Assignment

The Assignment tab displays the decision about of all units using the last induced rules.

Assignment table

For each unit, the table shows:

- s^- : lower class bound (best class/score supported by at-least rules)
- s^+ : upper class bound (worst class/score supported by at-most rules)
- **Assignment**: classification result
 - $s^- = s^+$: unit assigned to a single class
 - $s^- < s^+$: unit assigned to a range of classes (uncertain)
 - $s^- > s^+$: contradictory classification
- **Status**: OK or Contradictory

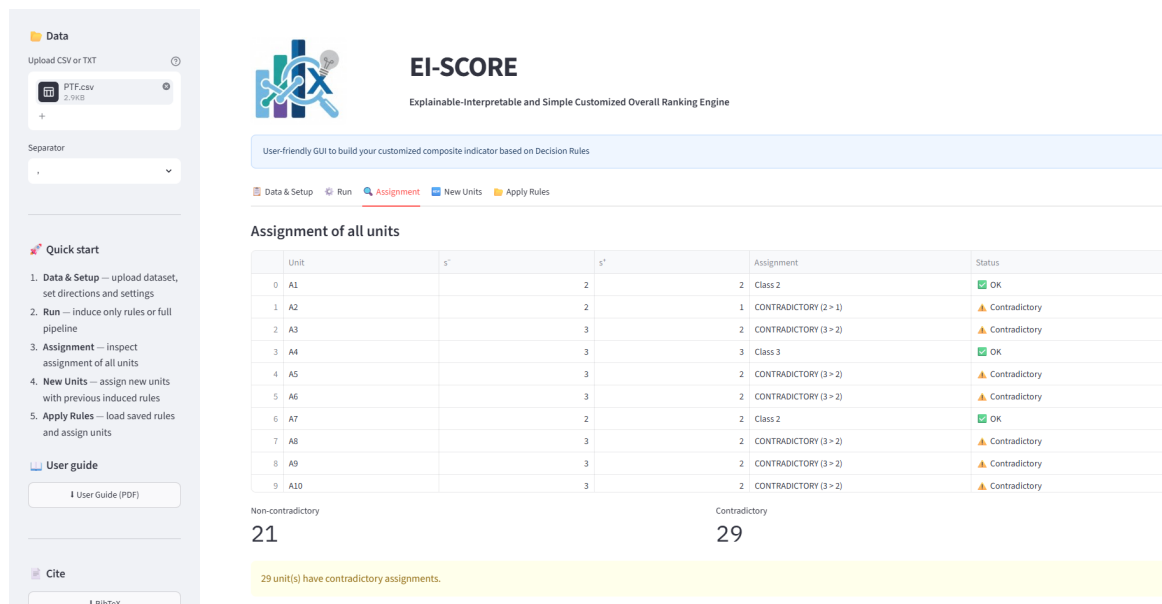


Figure 5: Tab 3 – Assignment table

Tip

Contradictory classification may appear only if you run *Rule induction only* in Tab 2 (Run).

Unit-by-unit explanation

Select a unit from the dropdown to see a detailed **explanation** of its classification/scoring: which rules were matched, and how s^- and s^+ were determined.

Unit-by-unit explanation

Select unit

k1

A1 → Class 2

Satisfied at-least rules:

r3: If $\text{Beta}(a) \geq 0.788$, then a is assigned to at least Class 2 ($d3 = 2$)

Satisfied at-most rules:

r2: If $\text{Net_Profit_Margin}(a) \leq 10.327$, then a is assigned to at most Class 2 ($d2 = 2$)

r3: If $\text{ROA}(a) \leq 4.94$, then a is assigned to at most Class 2 ($d3 = 2$)

r4: If $\text{EPS}(a) \leq 482.0$, then a is assigned to at most Class 2 ($d4 = 2$)

r5: If $\text{P_E}(a) \leq 9.156$, then a is assigned to at most Class 2 ($d5 = 2$)

r11: If $\text{Beta}(a) \leq 1.146$ and $\text{ROE}(a) \leq 22.855$, then a is assigned to at most Class 2 ($d11 = 2$)

r16: If $\text{Beta}(a) \leq 1.146$ and $\text{P_BV}(a) \leq 2.183$, then a is assigned to at most Class 2 ($d16 = 2$)

Download classification CSV

Figure 6: Tab 3 – Unit-by-unit explanation

A donwloadble CSV file is also available.

Tab 4 – New Units

The New Units tab allows classifying or scoring new units that were not part of the original dataset.

Uploading new units

Upload a CSV or TXT file containing the new units. The file must include the same criteria columns as the original dataset, **without** a class column. Unit names in the first column are optional. Once uploaded, the table of new units appears.

Data

Upload CSV or TXT

PTF.csv
3.9 KB

Separator

Quick start

1. **Data & Setup** — upload dataset, set directions and settings
2. **Run** — induce only rules or full pipeline
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5. **Apply Rules** — load saved rules and assign units

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EI-SCORE

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User-friendly GUI to build your customized composite indicator based on Decision Rules

Data & Setup Run Assignment **New Units** Apply Rules

Assign new units

Upload new units **without** the class column. The tool tries to handle contradictions.

Upload new units

PTF_new_units.csv
16.9 KB

Separator

	nome	Return	Beta	Net_Profit_Margin	ROA	ROE	EPS	P_E	P_BV
0	x1	-9.145	0.1	-10.159	-6.553	-47.223	-505.333	2.71	0.773
1	x2	66.644	1.1798	33.0816	17.0123	29.0545	1342.6334	28.2945	2.9225
2	x3	263.307	3.421	144.01	61.141	85.927	9331.333	250	20.786
3	x4	-9.145	0.1	144.01	61.141	85.927	9331.333	28.2945	2.9225

Assign new units

Figure 7: Tab 4 – New units loaded

Warning

The *full pipeline* must be run (Tab Run) before new units can be assigned to class or score.

Classification results

By clicking the button *Assign new units*, a set of maximal and minimum rules is found. The results table shows all original units together with the new units (highlighted in yellow), displaying:

- $s^-(prev)$: initial lower bound classification/score
- $s^+(prev)$: initial upper bound classification/score
- $s^-(new)$: final lower bound classification/score
- $s^+(new)$: final upper bound classification/score
- **Assignment**: final assignment
- **Changed**: whether the assignment changed from the initial due to the inclusion of new units

Assignment results

	Unit	$s^-(prev)$	$s^+(prev)$	$s^-(new)$	$s^+(new)$	Assignment	Changed
0	A1		2	2	2	2 Class 2	
1	A2		2	2	2	2 Class 2	
2	A3		3	3	2	3 Class 2 to 3	⚠ Yes
3	A4		3	3	3	3 Class 3	
4	A5		3	3	3	3 Class 3	
5	A6		2	2	2	2 Class 2	
6	A7		2	2	2	2 Class 2	
7	A8		3	3	3	3 Class 3	
8	A9		3	3	2	3 Class 2 to 3	⚠ Yes
9	A10		2	2	2	2 Class 2	

Changed in A

11

New units assigned

5

New contradictions

0

Summary result

Changed assignment of previous units: A3, A9, A14, A17, A22, A23, A24, A26, A44, A47, A49

$\mathcal{R}^{\geq/\leq}$ maximal rules selected: 56 at-least, 23 at-most

$\mathcal{R}^{\geq/\leq}$ minimal rules selected: 9 at-least, 5 at-most

Maximal rules selected (56 at-least, 23 at-most)(56 at-least, 23 at-most)

Minimal rules selected (9 at-least, 5 at-most)(9 at-least, 5 at-most)

Figure 8: Tab 4 – New units assignment results

A compact summary result is available with these information.

Unit-by-unit explanation

The unit-by-unit explanation is available for all units (original and new), using the minimal set of rules. Possibility to download units assignment, maximal and minimal set of rules.

Tab 5 – Apply Rules

The Apply Rules tab allows loading a previously saved rules file and applying it to units, without needing to re-run rule induction.

Tip

This tab is always accessible, even without loading a data file in the sidebar.

Loading rules

Upload a rules CSV file (in the format described in Section 3). The application automatically reads the `#directions` ($\uparrow$ or $\downarrow$) and `#mode` (class or score) lines to reconstruct the criterion directions and problem type.

Upon loading, the application displays:

- Number of at-least and at-most rules
- Criteria names and directions

Visualizing rules

The induced uploaded rules are displayed with the matching units highlighted (if alternatives have been loaded).

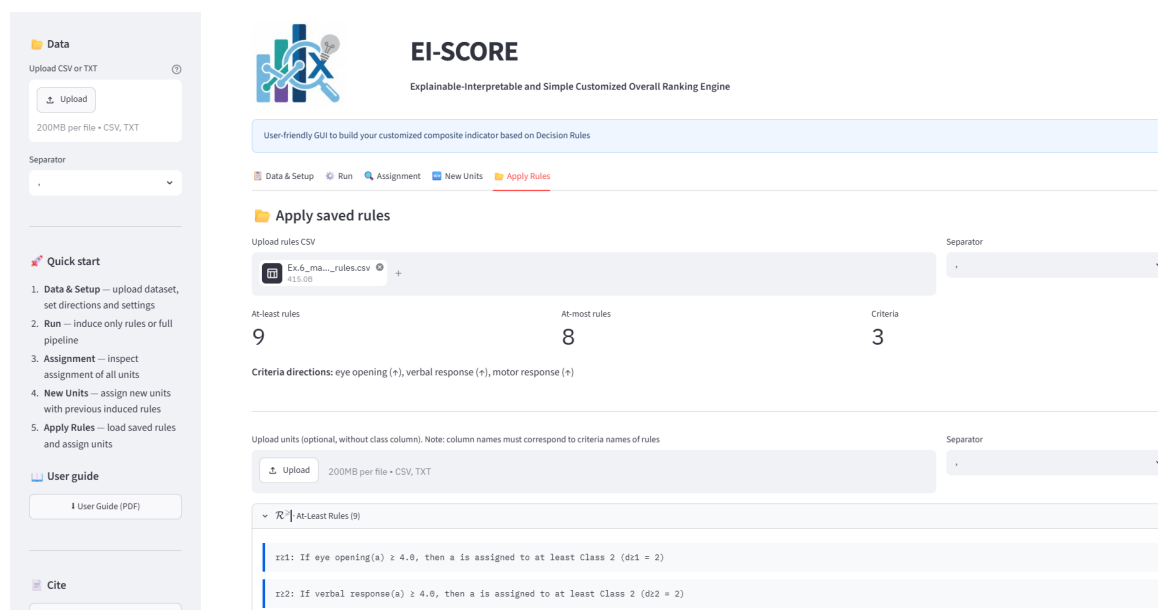


Figure 9: Tab 5 – Uploaded rules visualization

Loading units and view assignment

Optionally upload a file of units to assign them class or score using the loaded rules. The assignment table and unit-by-unit explanation are displayed.

Warning

The column names in the alternatives file must exactly match the criterion names in the rules file.

Assignment

	Unit	s^-	s^+	Assignment	Status
0	a1		1	1 Class 1	✓ OK
1	a2		2	2 Class 2	✓ OK
2	a3		2	2 Class 2	✓ OK
3	a4		1	1 Class 1	✓ OK
4	a5		1	1 Class 1	✓ OK
5	a6		2	2 Class 2	✓ OK
6	a7		3	3 Class 3	✓ OK
7	a8		3	3 Class 3	✓ OK
8	a9		2	2 Class 2	✓ OK

Non-contradictory

21

Contradictory

0

Unit-by-unit explanation

Select unit

a1

a1 → Class 1

No at-least rule satisfied → $s^- = 1$

Satisfied at-most rules:

rs1: If eye opening(a) ≤ 2.0, then a is assigned to at most Class 2 (ds1 = 2)

rs3: If verbal response(a) ≤ 2.0, then a is assigned to at most Class 1 (ds3 = 1)

rs4: If motor response(a) ≤ 4.0, then a is assigned to at most Class 2 (ds4 = 2)

Figure 10: Tab 5 – Assignment results from uploaded induced rules

Load a new unit dataset

After loading a set of rules and a set of units with non-contradictory assignments, one may investigate if the introduction of new units affect rules selection and all units assignments.

Assign new units

Upload new units to assign.

Upload new units (without class column)

unit	eye opening	verbal response	motor response
0 a22		2	4
1 a23		1	5

Assign new units

Done

Assignment results

	Unit	s^- (prev)	s^+ (prev)	s^- (new)	s^+ (new)	Assignment	Changed
13	a14		3	3	2	3 Class 2 to 3	▲ Yes
14	a15		1	1	1	1 Class 1	
15	a16		1	1	1	1 Class 1	
16	a17		2	2	2	2 Class 2	
17	a18		3	3	3	3 Class 3	
18	a19		2	2	2	2 Class 2	
19	a20		1	1	1	1 Class 1	
20	a21		3	3	3	3 Class 3	
21	a22		2	2	1	1 Class 1	■ New
22	a23		3	1	1	1 Class 1	■ New

Changed in A

2

New units assigned

2

New contradictions

0

Summary result

Maximal rules selected (8 at-least, 5 at-most)

Minimal rules selected (3 at-least, 3 at-most)

Figure 11: Tab 5 – Assignment results and rule selection after new units loaded

The shown information are similar to the one in Section 7.2. Moreover, unit-by-unit explanation,

CSV download are also available.

Tip

The result would be the same of running full pipeline in Run tab and, consequentially, tab New classification.

Examples

All example files are available in the **examples/** folder of the repository ([link](#)).

Example 1 – Human Development Index

Files: `examples/Ex.1_HDI.csv`

Problem: Classification

Dataset: 193 countries evaluated on 4 criteria (life expectancy, expected years of schooling, mean years of schooling, GNI per capita), assigned to 4 development classes.

Directions: all increasing (↑)

Workflow:

1. Sidebar: upload `Ex.1_HDI.csv`
2. Tab Data & Setup: set all criteria as increasing
3. Tab Run: select *Rule induction only* and click Run
4. Tab Run: inspect induced rules
5. Tab Assignment: inspect the countries assignment

Example 2 – Stock Portfolio

Files: `examples/Ex.2_PTF.csv`, `examples/Ex.2_PTF_new_units.csv`

Problem: Full pipeline + New units

Dataset: 50 stocks evaluated on 8 financial criteria (Return, Beta, Net Profit Margin, ROA, ROE, EPS, P/E, P/BV), assigned to 3 performance classes.

Directions: all increasing (↑)

Workflow:

1. Sidebar: upload `Ex.2_PTF.csv`
2. Tab Data & Setup: set all criteria as increasing
3. Tab Run: select *Full pipeline* and click Run
4. Tab Assignment: inspect stocks assignment
5. Tab New units: upload `Ex.2_PTF_new_units.csv` and assign new stocks
6. Tab New units: inspect any changes

Example 3 – Scored units

Files: `examples/Ex.3_ElectreScore.csv`

Problem: Scoring

Dataset: 10 investment projects evaluated on 5 criteria, ranked by a continuous score from 0 to 100.

Directions: g1 ↓, g2 ↓, g3 ↑, g4 ↑, g5 ↑

Workflow:

1. Sidebar: upload `Ex.3_ElectreScore.csv`
2. Tab Data & Setup: set criterion directions;
3. Tab Setup: select *Scoring* as problem type, select *Rule induction only* and click Run
4. Tab Setup: inspect rules – note that rules decision part shows original score values
5. Tab Assignment: inspect assignment – s^- and s^+ show score values

Example 4 – Missing Values

Files: `examples/Ex.4_MISSING.csv`, `examples/Ex.4_MISSING_new_units.csv`

Problem: Class assignment with missing values, Full pipeline

Dataset: 20 students evaluated on 3 criteria (mathematics score, reading score, socioeconomic index), assigned to 3 performance classes. Several missing values are present.

Directions: all increasing ($\uparrow$)

Workflow:

1. Sidebar: upload `Ex.4_MISSING.csv`
2. Tab Data & Setup: inspect NaN/missing values are correctly recognized
3. Tab Run: select *Full pipeline* and click Run
4. Tab Assignment: inspect class assignment
5. Tab New units: upload `Ex.4_MISSING_new_units.csv` and assign class to new students

Example 5 – Apply Rules

Files: `examples/Ex.5_GCS_original.csv`, `examples/Ex.5_GCS_new_units.csv`, `examples/Ex.5_maximal_rules.csv`

Problem: Load previous induced rules, assign units, assign new units

Dataset: Pre-computed rules for the GCS dataset (9 at-least, 8 at-most rules, 3 criteria, class mode).

Workflow:

1. Tab Apply rules: upload `Ex.5_maximal_rules.csv` and inspect the upload rules
2. Tab Apply rules: upload `Ex.5_GCS_original.csv` and inspect their class assignment – s^- and s^+
3. Tab Apply rules: upload `Ex.5_GCS_new_units.csv` and run *Assign new units*
4. Tab Apply rules: inspect new units assignment

Troubleshooting**File format errors**

- **“Could not read file”:** the file separator does not match the selected separator. Try changing the separator dropdown (comma, semicolon, tab, space).
- **Criteria columns show unexpected values:** ensure that the class/score label is in the last column of the file. All intermediate columns are treated as criteria.

- **Non-numeric values in criteria:** columns with text values are treated as unit names (first column) or ignored. Ensure all criteria columns contain numeric values only.
- **No reference units detected:** if all class/score labels are empty, no rules can be induced. Ensure that at least some units have a numeric entry in the last column.

Rules file errors

- **“Missing #directions line”:** the rules file must start with a `#directions` line. Check the file format in Section 3.
- **Column names do not match** (Tab Apply rules): when classifying alternatives in Tab Apply rules, the column names in the units file must exactly match the criterion names in the rules file (case-sensitive, no extra spaces).
- **Wrong #mode:** if rules were exported from a scoring problem, the file contains `#mode,score`. Loading it in Tab Apply rules will display score values instead of class indices. This is correct behavior.

Pipeline errors

- **“Run the Full pipeline first”** (Tab New units): the New Units tab requires the full pipeline to be completed first. Go to Tab Run, select *Full pipeline*, and click Run.
- **Rule induction takes a long time:** for large datasets with many contradictions, the rule induction may take several minutes. This is expected behavior.

How to Cite

If you use this software in your research, projects, or publications, please cite the following papers.

- Corrente, S., Greco, S., Słowiński, R., & Zappalà, S. (2026). *An explainable and interpretable composite indicator based on decision rules*. Omega, 142, 103513. <https://doi.org/10.1016/j.omega.2026.103513>
- Greco, S., Matarazzo, B., & Slowinski, R. (2001). *Rough sets theory for multi-criteria decision analysis*. European Journal of Operational Research, 129(1), 1–47. [https://doi.org/10.1016/S0377-2217\(00\)00167-3](https://doi.org/10.1016/S0377-2217(00)00167-3)

For BibTeX users you can cite them directly in your L^AT_EX document using `\cite{corrente2026}` and `\cite{greco2001}`:

```
@article{corrente2026,
  title      = {An explainable and interpretable composite indicator
               based on decision rules},
  author     = {Corrente, Salvatore and Greco, Salvatore and S{\l}owi
               {\n}ski, Roman and Zappal{\a}, Silvano},
  journal    = {Omega},
  volume     = {142},
  pages      = {103513},
  year       = {2026},
  publisher  = {Elsevier},
  doi        = {10.1016/j.omega.2026.103513}
}
```



```
@article{greco2001,  
  title      = {Rough sets theory for multicriteria decision analysis  
               },  
  author     = {Greco, Salvatore and Matarazzo, Benedetto and  
               Slowinski, Roman},  
  journal    = {European journal of operational research},  
  volume     = {129},  
  number     = {1},  
  pages      = {1--47},  
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